Advanced Computer Networks – PCS223

Assignment 7

Name – Writick Parui
Subgroup – CS4
Roll No. – 8025320111

Objective:

To configure and verify OSPF routing and analyze SPF algorithm.

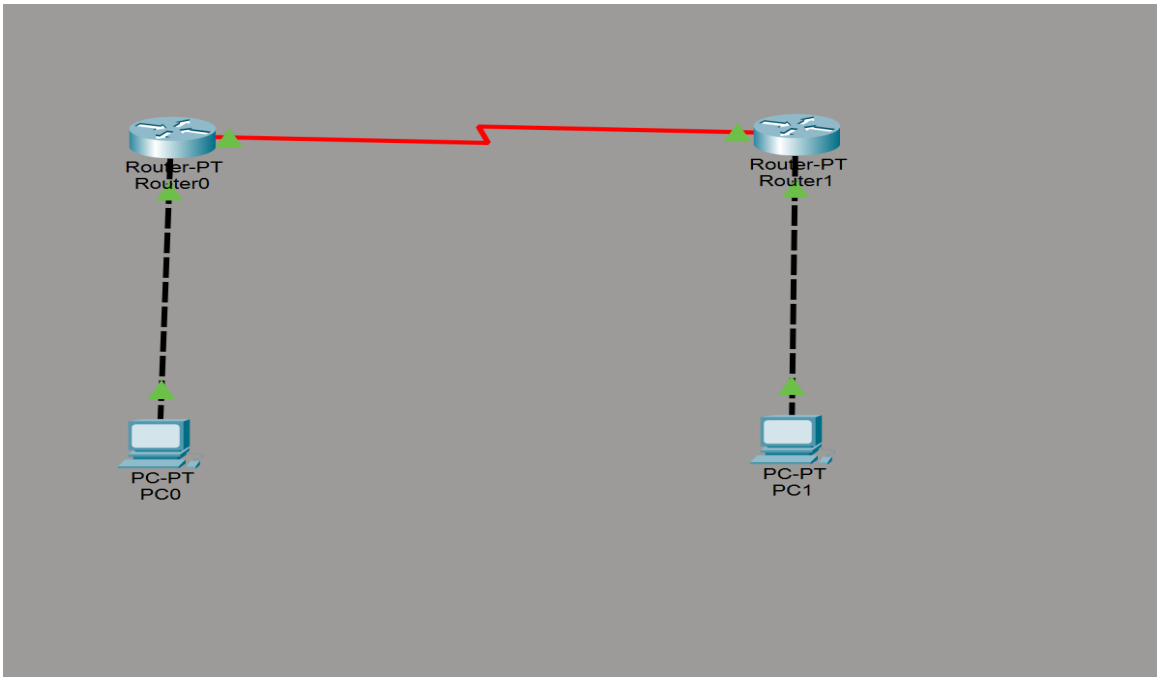
Router Configuration:

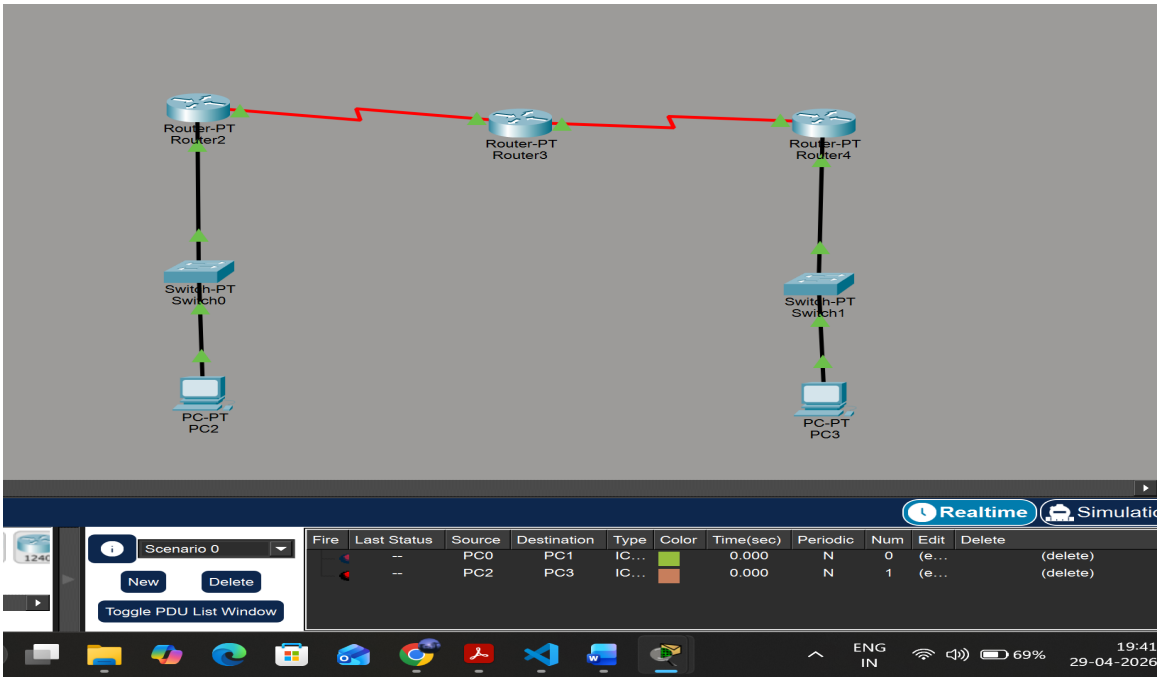
```
R1:
router ospf 1
router-id 1.1.1.1
network 100.1.1.0 0.0.0.255 area 0
network 20.1.1.0 0.0.0.255 area 0

R2:
router ospf 1
router-id 2.2.2.2
network 100.1.1.0 0.0.0.255 area 0
network 172.16.0.0 0.0.255.255 area 0

R3:
router ospf 1
router-id 3.3.3.3
network 20.1.1.0 0.0.0.255 area 0
network 10.2.2.0 0.0.0.255 area 0
```

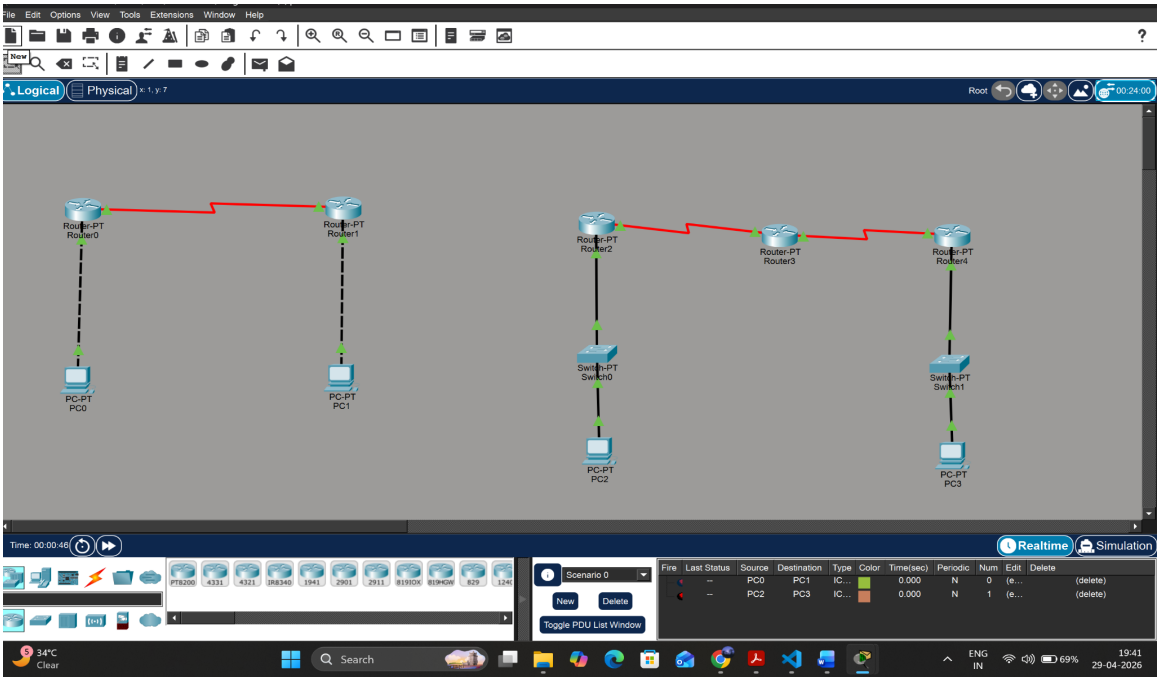
Verification Commands:

- show ip route
- show ip ospf neighbor
- show ip ospf database
- ping between PCs successful

Output Screenshots:


Router-PT
Router0
Router-PT
Router1
PC-PT
PC0
PC-PT
PC1


Router-PT
Router2
Router-PT
Router3
Router-PT
Router4
Switch-PT
Switch0
Switch-PT
Switch1
PC-PT
PC2
PC-PT
PC3
Realtime
Simulatio
Scenario 0
New
Delete
Toggle PDU List Window
Fire | Last Status | Source | Destination | Type | Color | Time(sec) | Periodic | Num | Edit | Delete
-- | PC0 | PC1 | IC... | 0.000 | N | 0 | (e... | (delete)
-- | PC2 | PC3 | IC... | 0.000 | N | 1 | (e... | (delete)
ENG
IN
69%
19:41
29-04-2026



Conclusion:

OSPF routing successfully configured. All routers formed adjacency and routes were learned dynamically. Connectivity verified between all PCs.